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Shot-Based Quantum Encoding: A Data-Loading Paradigm for Quantum Neural Networks

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ABSTRACT

Efficient data loading remains a bottleneck for near-term quantum machine learning. Existing schemes (angle, amplitude, and basis encoding) either underuse the exponential Hilbert-space capacity or require circuit depths that exceed the coherence budgets of noisy intermediate-scale quantum hardware. We introduce shot-based quantum encoding (SBQE), a data embedding strategy that distributes the hardware's native resource, shots, according to a data-dependent classical distribution over multiple initial quantum states. By treating the shot counts as a learnable degree of freedom, SBQE produces a mixed-state representation whose expectation values are linear in the classical probabilities and can therefore be composed with nonlinear activation functions. We show that SBQE is structurally equivalent to a multilayer perceptron whose weights are realized by quantum circuits, and we describe a hardware-compatible implementation protocol. Benchmarks on three image datasets, with 10 independent initializations per model, show that SBQE achieves $89.1\% \pm 0.9\%$ test accuracy on Semeion (reducing error by 5.3% relative to amplitude encoding and matching a width-matched classical network), $80.95\% \pm 0.10\%$ on Fashion MNIST (exceeding amplitude encoding by +2.0% and a linear multilayer perceptron by +1.3%), and $90.25\% \pm 0.18\%$ on MNIST (exceeding amplitude encoding by 2.1 percentage points and the width-matched classical network by 0.3), all without any data-encoding gates.

1 | Introduction

Quantum machine learning (QML) seeks to combine superposition, entanglement, and the exponential dimensionality of Hilbert space with classical learning algorithms [1–3]. Early results point to possible advantages in kernel evaluation [4, 5], generative modeling [6, 7], and combinatorial optimization and routing [8–10]. Yet two practical hurdles still limit what can be achieved with noisy-intermediate-scale quantum (NISQ) devices: (i) *encoding* classical data efficiently into a quantum register, and (ii) *extracting* task-relevant information through inherently noisy, sample-limited measurements. The present work addresses the first challenge.

Current data-loading paradigms fall into two main families. *Angle encoding* assigns one feature to one qubit by mapping the feature value onto a single-qubit rotation [11, 12]. This strategy benefits from shallow circuits and straightforward hardware implementation, but its representational capacity scales only linearly with the number of qubits, leaving the exponential Hilbert-space dimension largely untapped. Data-reuploading techniques partially mitigate this limitation by inserting additional encoding layers throughout the variational circuit, at the cost of greater depth and noise susceptibility [13, 14].

Amplitude encoding, in contrast, loads up to 2^n classical amplitudes into an n -qubit state, thereby saturating the available

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quantum degrees of freedom [15]. Unfortunately, exact state-preparation routines typically require $\mathcal{O}(2^n)$ controlled rotations and entangling gates, which exceed the coherence and fidelity budgets of current hardware [16].

We introduce shot-based quantum encoding (SBQE), which bypasses the depth bottleneck by moving the encoding workload from coherent gates to the classical control layer that orchestrates repeated circuit executions (shots). Instead of preparing a single pure state per datum, SBQE assigns each data point a classical probability vector that dictates how many shots start from each member of a fixed, hardware-friendly set of basis states. The resulting statistical mixture constitutes the input to a shallow variational circuit; expectation values remain linear in the classical probabilities, so they compose directly with nonlinear classical post-processing and multilayer QML architectures. Because contemporary NISQ experiments already require thousands of repetitions to estimate observables with meaningful precision [17], SBQE exploits an abundant resource without increasing coherent depth.

The remainder of this paper is organized as follows. Section 2 reviews competing data-loading strategies and situates SBQE within the broader QML literature. Section 3 presents the density-matrix formulations of existing encodings, introduces the SBQE framework, and establishes its connection to multilayer perceptrons. Numerical benchmarks on three image-classification tasks are presented in Section 4, and Sections 5 and 6 discuss broader implications and limitations.

2 | Background and Related Work

Variational quantum circuits (VQCs) underpin most algorithms designed for the NISQ era. A VQC applies a parameterized unitary $U(\theta)$ to an initial state, measures a set of observables, and iteratively updates the angles θ to minimize a classical loss function [2, 11, 18]. Early exemplars such as the Variational Quantum Eigensolver and the Quantum Approximate Optimisation Algorithm [8] validated the hybrid optimization loop, while subsequent studies analyzed gradient landscapes and identified the barren-plateau issue, namely an exponential decay of trainable signal with qubit count [19]. Strategies including layer-wise training, problem-inspired ansätze, and noise-aware optimizers now aim to mitigate such pathologies.

The expressive power of any VQC ultimately depends on how classical information is embedded into the quantum register. In *angle encoding* a real-valued feature x modulates a single-qubit rotation, typically an $R_y(x)$ or $R_z(x)$ gate. This linear mapping scales only with the number of qubits but enjoys shallow depth and straightforward hardware execution [4, 12]. Depth-amplified variants, sometimes called “quantum depth-infused” (QDI) encodings, repeatedly interleave feature-dependent rotations with entangling layers to enrich the effective feature map [13], and have been deployed in industrial prediction and control settings [20]. By contrast, *amplitude encoding* prepares a state whose computational-basis amplitudes store the entire data vector, achieving an exponential compression from 2^n features into n qubits [15]. Exact state preparation, however, requires gate counts that scale exponentially in the worst case; schemes based

on uniformly controlled rotations [21], isometry decomposition [22], or quantum random access memory (QRAM) access [23] alleviate but do not eliminate the depth overhead.

An alternative line of research embraces *probabilistic and mixed-state* descriptions. Instead of loading information coherently, one prepares a classical or noisy ensemble of pure states whose density matrix encodes the data. Such mixtures can bypass barren plateaus, exploit naturally occurring noise, or reduce coherent depth [24]. Sampling-based protocols that randomize over easy-to-prepare states have been proposed for fair sampling and error mitigation [25], while classical-shadow methods treat measurement randomness itself as a computational resource [26]. The present work builds on this perspective by turning the classical allocation of circuit repetitions (shots) into a learnable data-dependent degree of freedom, thereby unifying probabilistic encodings with the trainable expressivity of VQCs.

3 | Shot-Based Quantum Encoding

3.1 | Encoding Schemes in the Density-Matrix Picture

The output of a variational quantum circuit can be written in full generality as

$$f(\mathbf{x}, \theta) = \text{Tr}[O \rho(\mathbf{x}, \theta)], \quad (1)$$

where ρ is a density matrix and O is a Hermitian observable. Both ρ and O may depend on inputs $\mathbf{x}$ and trainable parameters θ [15, 27]. The encoding strategy determines how ρ is constructed from a reference state $|0\rangle$.

Basis encoding maps an integer $i \in \{0, \dots, 2^n - 1\}$ to the corresponding computational-basis state by flipping individual qubits with Pauli- X gates

$$\rho(i, \theta) = U(\theta) |i\rangle\langle i| U^\dagger(\theta), \quad i \in \mathbb{N}. \quad (2)$$

Angle encoding embeds continuous features through single-qubit rotations, making the unitary itself data-dependent. Because the rotations act at the individual-qubit level, the number of encodable features scales linearly with the number of qubits, in contrast to basis, amplitude, and shot-based encoding, all of which can accommodate exponentially more features than qubits [12, 28]

$$\rho(\mathbf{x}, \theta) = U(\mathbf{x}, \theta) |0\rangle\langle 0| U^\dagger(\mathbf{x}, \theta). \quad (3)$$

Amplitude encoding places an L^2 -normalized data vector directly into the probability amplitudes of the state, compressing up to 2^n features into n qubits at the cost of an exponentially deep preparation circuit [15]

$$\rho(\mathbf{x}, \theta) = U(\theta) |\mathbf{x}\rangle\langle \mathbf{x}| U^\dagger(\theta). \quad (4)$$

All three schemes above produce *pure* states. SBQE departs from this pattern by encoding data into a *mixed* state whose mixture weights are themselves the data representation.

3.2 | Density-Matrix Formulation of SBQE

Let $\{|\psi_j\rangle\}_{j=1}^n$ be a fixed, efficiently preparable set of pure states on q qubits. SBQE maps a datum $\mathbf{x}$ to a probability vector $\mathbf{p}(\mathbf{x}) = (p_1, \dots, p_n)$ with $\sum_j p_j = 1$. Executing N_{tot} shots, we draw integer counts $\mathbf{N}(\mathbf{x}) = (N_1, \dots, N_n)$ from Multinomial($N_{\text{tot}}, \mathbf{p}(\mathbf{x})$) and initialize exactly N_j runs in $|\psi_j\rangle$. The encoded state is therefore the classical mixture

$$\rho(\mathbf{x}) = \sum_{j=1}^n p_j(\mathbf{x}) |\psi_j\rangle\langle\psi_j|. \quad (5)$$

Subsequent application of a variational unitary $U(\theta)$ and measurement of Hermitian observables $\{O_i\}_{i=1}^m$ yields outputs

$$f_i(\mathbf{x}, \theta) = \text{Tr}[O_i U(\theta) \rho(\mathbf{x}) U^\dagger(\theta)]. \quad (6)$$

As a concrete example, consider a single-qubit register with two initial states $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$ and a total budget of $N_{\text{tot}} = 1024$ shots. For a datum $\mathbf{x}$ whose probability vector is $\mathbf{p}(\mathbf{x}) = (2/3, 1/3)$, the experiment allocates $\mathbf{N} = (683, 341)$ shots to $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$, respectively. If the initial-state pool is enlarged to $\{|0\rangle\langle 0|, |1\rangle\langle 1|, |+\rangle\langle +|, |-\rangle\langle -|\}$, the probability vector has four components, for example, $\mathbf{p} = (3/8, 1/4, 1/4, 1/8)$, giving $\mathbf{N} = (384, 256, 256, 128)$. This contrasts with the standard practice of starting all 1024 shots in $|0\rangle\langle 0|$; here the data are encoded entirely through the classical redistribution of shots, without any additional quantum gates.

When the initial states are the 2^q computational-basis states and the observables are projective measurements in the same basis, Equation (6) simplifies to $f_i = |U_{ij}(\theta)|^2 p_j(\mathbf{x})$, so that $\mathbf{f}$ is itself a probability vector that can be fed directly into a subsequent SBQE layer.

In the experiments reported below, the probability vector $\mathbf{p}(\mathbf{x})$ is not fixed in advance but is produced by a trainable map applied to the reduced data, so that the encoding is adapted to the task rather than prescribed by a fixed pixel-to-probability rule. Each datum is first reduced to its $d = 8$ leading principal components $\mathbf{x} \in \mathbb{R}^d$ (Section 4) and rescaled to the unit interval; a trainable affine layer maps it to $\mathbf{z} = W\mathbf{x} + \mathbf{b} \in \mathbb{R}^n$, and an element-wise absolute value followed by ℓ_1 normalization projects $\mathbf{z}$ onto the probability simplex,

$$p_j(\mathbf{x}) = \frac{|z_j|}{\sum_k |z_k|}, \quad p_j \geq 0, \quad \sum_j p_j = 1. \quad (7)$$

The weights W and biases $\mathbf{b}$ are optimized jointly with the circuit parameters θ by back-propagation. Taking the pool $\{|\psi_j\rangle\}$ to be the full computational basis, so that $n = 2^q$, the entries of $\mathbf{p}(\mathbf{x})$ are the populations of the 2^q basis states and the encoded state is diagonal in that basis, $\rho(\mathbf{x}) = \sum_j p_j(\mathbf{x}) |j\rangle\langle j| = \text{diag}(\mathbf{p}(\mathbf{x}))$; it is realized physically by allocating a fraction p_j of the N_{tot} shots to $|j\rangle$ (prepared with single-qubit X gates), with no data-dependent rotation or entangling gates. Figure 1 illustrates this pipeline on two trained Fashion-MNIST examples.

3.3 | Connection to Multilayer Perceptrons

Inserting Equation (5) into Equation (6),

$$f_i(\mathbf{x}, \theta) = \sum_{j=1}^n W_{ij}(\theta) p_j(\mathbf{x})$$

$$W_{ij}(\theta) := \langle \psi_j | U^\dagger(\theta) O_i U(\theta) | \psi_j \rangle. \quad (8)$$

Hence each layer is a *linear* map in the encoded probabilities, analogous to the affine transformation in classical MLPs [29]. Placing the classical and quantum cases side by side makes the correspondence explicit

$$\text{Classical: } n_i = \sigma(\widetilde{W}_{ij}(\theta) x_j), \quad (9)$$

$$\text{Quantum: } n_i = \sigma(W_{ij}(\theta) p_j(\mathbf{x})). \quad (10)$$

The key differences are that in the quantum case the data must first be processed into a probability vector $\mathbf{p}(\mathbf{x})$, and that $\widetilde{W}_{ij}$ stores trainable weights directly, whereas W_{ij} is the output of a quantum device and depends on θ only indirectly through $U(\theta)$.

A nonlinearity σ that maps its input back to a valid probability vector allows the output of one layer to serve as the shot distribution for the next, giving a multilayer architecture

$$\mathbf{p}^{(k+1)} = \sigma(W^{(k)}(\theta^{(k)}) \mathbf{p}^{(k)}), \quad (11)$$

with $\mathbf{p}^{(0)} \equiv \mathbf{p}(\mathbf{x})$. SBQE therefore supports stacked “quantum perceptrons” in which *all* trainable weights reside in quantum parameters θ .

One such activation function that preserves the probability-vector structure is the *log-ReLU cascade*

$$\begin{aligned} \mathbf{p}' &= \log \mathbf{p}, \\ \mathbf{p}'' &= \mathbf{p}' - \langle \mathbf{p}' \rangle, \\ \mathbf{p}''' &= \text{ReLU}(\mathbf{p}''), \\ \sigma(\mathbf{p}) &= \mathbf{p}''' / \sum_j p_j'''. \end{aligned} \quad (12)$$

Two edge cases require special treatment: if $p_i = 0$, the entry is set to zero and excluded from the remaining steps; if $\mathbf{p}''' = \mathbf{0}$, the output defaults to a uniform distribution over the surviving entries. This function zeros out low-probability components, which is well suited for finite-shot experiments where small probabilities cannot be resolved anyway.

4 | Numerical Experiments

4.1 | Benchmarks

We evaluate SBQE in a machine learning setting on three widely used image-classification suites that differ in visual complexity and class balance. *Fashion-MNIST* offers 70 000 28×28 grayscale product images drawn from 10 garment categories [30]. *Semeion*

Encoding a Fashion-MNIST image into a mixed state via the learned probability map (8 qubits, trained model)
 $\text{image} \rightarrow 8 \text{ PCA features } \mathbf{x} \rightarrow p(\mathbf{x}) = |W\mathbf{x} + b|/\sum |W\mathbf{x} + b| \rightarrow \rho(\mathbf{x}) = \text{diag}(p)$

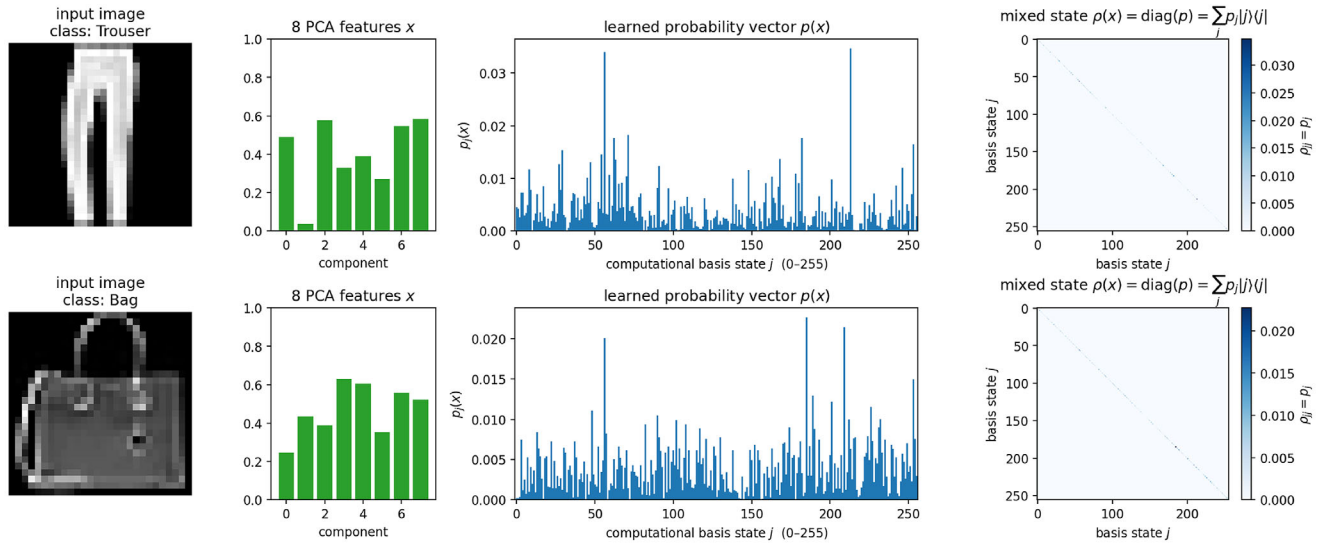


FIGURE 1 | Encoding a datum into a mixed state through the learned probability map ($q = 8$ qubits, trained model). For two Fashion-MNIST images of different classes (a trouser and a bag), from left to right: the input image; its $d = 8$ principal-component features $\mathbf{x}$; the learned probability vector $\mathbf{p}(\mathbf{x})$ over the $2^8 = 256$ computational-basis states Equation (7); and the induced mixed state $\rho(\mathbf{x}) = \text{diag}(\mathbf{p}(\mathbf{x})) = \sum_j p_j |j\rangle\langle j|$, which is diagonal in the computational basis. Images of different classes yield distinct probability vectors, and hence distinct mixed states.

comprises 1593 handwritten digits collected from roughly 80 individuals and pre-binarized to 256 input pixels [31]. *MNIST* comprises 70 000 28×28 grayscale images of handwritten digits [32]. All three tasks are cast as 10-class classification.

Each raw image is vectorized, centered, and projected onto the first $d = 8$ principal components via PCA, reducing the dimensionality to match an eight-qubit register. Principal-component analysis [33] is implemented with *scikit-learn*'s exact SVD routine (double precision). The same $d = 8$ reduction and eight-qubit register are used for MNIST as for the other datasets; this is distinct from the illustrative 784-pixel, 10-qubit configuration mentioned in Section 5.

4.2 | Experimental Setup

4.2.1 | Model Zoo

We compare three architectures that share the same quantum workload ($n_q = 8$ qubits, $n_{\text{layers}} = 4$) and are matched for *total trainable parameters* wherever applicable.

1. *Amplitude Hybrid*: Baseline with deterministic amplitude embedding followed by variational layers with Rot-CNOT ladder, identical to the template in [34].
2. *SBQE Hybrid*: Our SBQE variant: an input dense layer produces a simplex-normalized probability vector, which is fed to the same variational circuit as above.
3. *Width-Matched Linear*: A two-layer classical MLP whose hidden width is computed analytically so that its parameter count matches that of the quantum hybrids as closely as possible without exceeding it.

4.2.2 | Circuit Details

All quantum layers employ the $\text{Rot}(\alpha, \beta, \gamma)$ gate per qubit followed by linear-chain CNOTs; parameters are initialized from $\mathcal{N}(0, 0.02)$. Statevector back-propagation is provided by PennyLane 0.41 [34] and executed on a single NVIDIA A100 GPU via JAX 0.4.28 [35]; comparable experiments can be reproduced with the TQml simulator [36]. Classical submodules are written in Flax 0.8.

4.2.3 | Optimization and Regularization

All models minimize the cross-entropy loss with the Lion optimizer ($\beta_1 = 0.95, \beta_2 = 0.98$, decoupled weight decay 10^{-2} ; batch size 7000) over 20 000 epochs. Gradients are clipped to a global norm of 0.01.

4.2.4 | Training Protocol

For each dataset, we draw stratified splits of 70%/15%/15% for training, validation, and testing, fixing the random seed across baselines. Every configuration is run 10 times with independent parameter initialization, and metrics are averaged. Early stopping is *disabled* in order to expose asymptotic behavior; instead we report the best validation-epoch weights on the held-out test set.

4.3 | Results

We trained each architecture described in Section 4 for 10 independent random initializations and report test-set performance as mean $\pm$ standard deviation. Convergence trajectories are shown

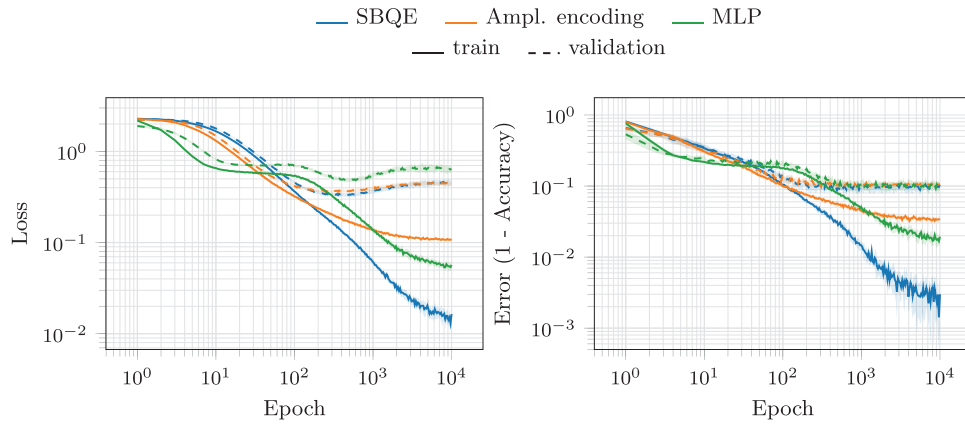


FIGURE 2 | Learning curves on the SEMEION benchmark (8 qubits, 4 layers). Left panel: cross-entropy loss; right panel: classification error (1 – accuracy). Within each panel, solid lines are training and dashed lines validation, both averaged over 10 runs; shaded bands show one standard deviation, and colors distinguish the three models as in the legend.

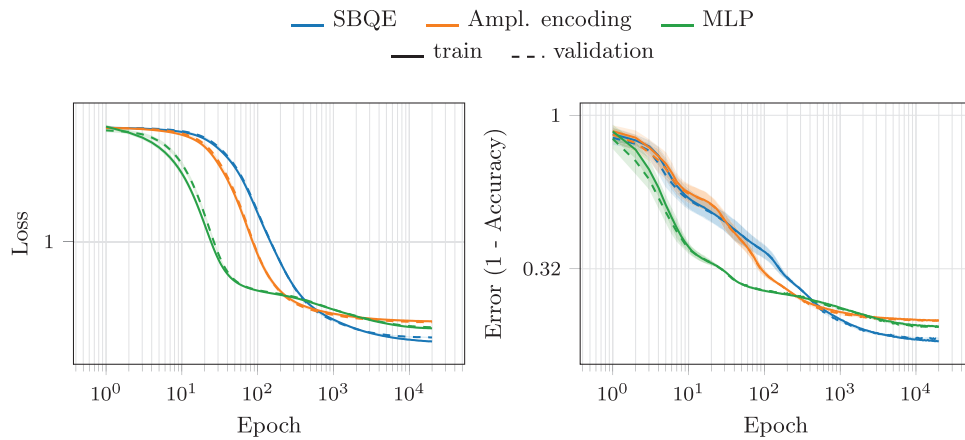


FIGURE 3 | Learning curves on the FASHION-MNIST benchmark (8 qubits, 4 layers). Left panel: cross-entropy loss; right panel: classification error (1 – accuracy). Within each panel, solid lines are training and dashed lines validation, both averaged over 10 runs; shaded bands show one standard deviation, and colors distinguish the three models as in the legend.

in Figure 2 (Semeion), Figure 3 (Fashion-MNIST), and Figure 4 (MNIST).

4.3.1 | Semeion Handwritten Digits

SBQE attains a test accuracy of $89.1\% \pm 0.9\%$ and a cross-entropy loss of 0.428 ± 0.037 . This represents a relative error reduction of 5.3% against amplitude encoding at identical depth and qubit count, and narrows the gap to a width-matched classical MLP ($89.6\% \pm 1.3\%$) to within statistical noise. Although the state-of-the-art CNNs exceed 95% on this dataset [37], such models employ $> 10^5$ parameters and convolutional inductive bias, whereas all three baselines here are capped below 1.1×10^4 parameters. The result therefore demonstrates that reallocating shot resources already available on NISQ hardware can recover essentially all of the performance given by purely classical depth.

4.3.2 | Fashion-MNIST

On the harder Fashion benchmark, SBQE again leads with $80.95\% \pm 0.10\%$ accuracy, outperforming amplitude encoding

by an absolute 2.0% and the linear MLP by 1.3%. On the same eight PCA features a purely linear classifier (multinomial logistic regression) reaches only 73.5%, so SBQE adds 7.5 percentage points on identical inputs, whereas amplitude-encoded quantum classifiers reported in the literature rely on deeper data-reuploading circuits and achieve 78%–79% at eight qubits [38, 39]. A linear classifier reaches 85% only when given the full 784-pixel input rather than the eight components used here [40].

4.3.3 | MNIST

On the standard handwritten-digit benchmark, SBQE reaches $90.25\% \pm 0.18\%$ accuracy, again the highest of the three models, exceeding amplitude encoding $88.17\% \pm 0.09\%$ by 2.1 percentage points and the width-matched classical network $89.95\% \pm 0.14\%$ by 0.3. Convergence curves are shown in Figure 4. Evaluated under the identical protocol used for the other datasets, this confirms that the advantage of shot-based mixing over amplitude encoding is not specific to the smaller Semeion and Fashion suites.

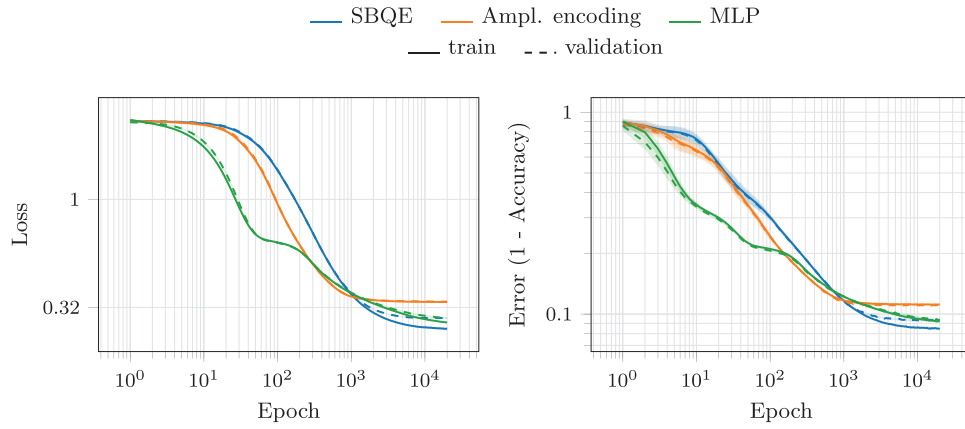


FIGURE 4 | Learning curves on the MNIST benchmark (8 qubits, 4 layers). Left panel: cross-entropy loss; right panel: classification error ($1 - \text{accuracy}$). Within each panel, solid lines are training and dashed lines validation, both averaged over 10 runs; shaded bands show one standard deviation, and colors distinguish the three models as in the legend.

4.3.4 | Statistical Comparison

Two-tailed paired t -tests over the 10 seeds confirm that SBQE is significantly better than amplitude encoding on all three datasets ($p < 0.01$ for Semeion and Fashion-MNIST, $p \approx 2 \times 10^{-10}$ for MNIST); against the width-matched linear network the difference is not significant for Semeion but is for Fashion-MNIST ($p < 0.05$) and MNIST ($p \approx 2 \times 10^{-3}$). Taken together, the results support our central hypothesis that shot allocation is a competitive information channel for near-term quantum classifiers.

4.4 | Contribution of the Quantum Layer

A natural question is whether the advantage of SBQE stems from the quantum circuit itself or merely from the trainable classical layer that maps the reduced features to $\mathbf{p}(\mathbf{x})$. To isolate the circuit's contribution, we replace it with a same-shape classical map of $\mathbf{p}(\mathbf{x})$, a trainable linear layer $2^{n_q} \rightarrow n_q$, and sweep its rank, holding the SBQE front-end and readout fixed. As shown in Figure 5, the quantum layer, whose mixing stage carries only $3(n_{\text{layers}} + 1)n_q = 120$ trainable angles, matches or exceeds classical maps that use an order of magnitude more parameters; at the quantum layer's own parameter budget the classical stand-in reaches only 57%–67%. The quantum circuit, therefore, supplies representational capacity beyond that of the classical front-end alone, confirming that the separation is driven by the circuit rather than by the preprocessing.

5 | Discussion

SBQE uses an overlooked resource, *classically allocated shots*, to sidestep the data-loading bottleneck. By eliminating encoding gates, we reduce both depth and coherent error accumulation, rendering the scheme compatible with today's ~ 100 -gate error budgets. Further, SBQE meshes naturally with measurement-based and classical-shadow post-processing; extending the framework to error-mitigated estimators is a promising direction. Noise-induced regularization techniques [41] could be applied on top of SBQE to further improve generalization on noisy hardware. Likewise, information-plane analysis, which quantify how

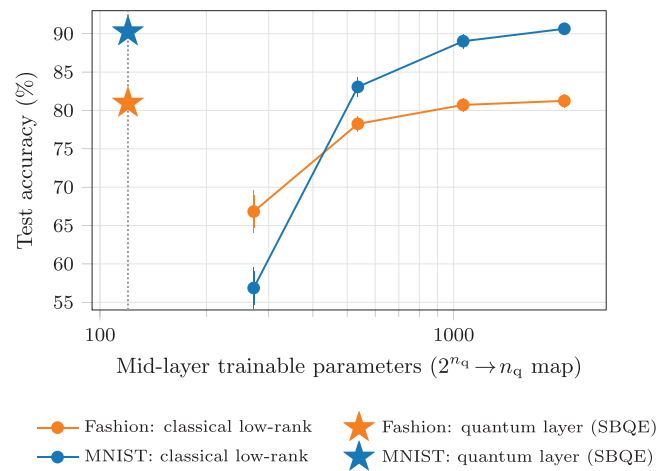


FIGURE 5 | Test accuracy of the SBQE quantum layer (stars) compared with same-shape classical low-rank maps of $\mathbf{p}(\mathbf{x})$ ($2^{n_q} \rightarrow n_q$) as a function of the mixing layer's trainable-parameter count ($n_q = 8$ qubits, 4 layers; 10 seeds, best-validation weights). The quantum layer reaches its accuracy with far fewer parameters than a classical map of comparable performance, indicating that the circuit (not the front-end or the PCA reduction) carries the class separation.

strongly a model compresses its input and feed this signal back into training [42], offer a natural tool for monitoring and tuning the learned probability map $p(\mathbf{x})$.

Another point concerns linearly separable data. Standard variational quantum circuits with angle encoding produce truncated Fourier series and are known to struggle with linear decision boundaries [12, 28, 43, 44]. Because SBQE encodes data through classical probabilities rather than through rotation angles, the resulting model class is not restricted to periodic functions, and preliminary experiments on synthetic two-dimensional tasks confirm that SBQE can separate linear data with ease.

Beyond the image benchmarks presented here, SBQE is well suited to high-dimensional inputs. Since each initial state in the pool corresponds to one component of the probability vector, a register of q qubits with 2^q basis states can encode up to 2^q

features without any encoding gates. For instance, the 784 pixels of an MNIST image fit into a 10-qubit first layer ($784 < 2^{10} = 1024$). This logarithmic scaling of qubits with feature count is shared with amplitude encoding but achieved here without the associated circuit depth.

Limitations. (i) Shot budgets grow with feature dimension if one insists on low-variance multinomial sampling; at a minimum, the total shot count must be large enough that each datum produces a distinguishable distribution. How expressivity degrades as the shot budget decreases is an important open question. Stratified resampling strategies may help. (ii) The preprocessing step requires choosing an appropriate probability mapping $\mathbf{p}(\mathbf{x})$ for each dataset. Functions such as softmax introduce their own nonlinearity, which is useful given that the quantum layer itself is linear in the probabilities, but a more principled selection method would be valuable. (iii) Hardware constraints on state-preparation variety $\{|\psi_j\rangle\}$ could limit expressivity; adaptive state pools deserve exploration. Superposition of parameterized circuits [45] is one route toward richer initial-state families without proportional depth overhead. (iv) Multilayer SBQE requires multiple sequential passes through the quantum device, which can be costly in wall-clock time.

6 | Conclusion

We have introduced SBQE, a data-loading method that replaces gate-heavy encodings with probabilistic shot allocations. The resulting mixed-state formalism is structurally equivalent to a classical multilayer perceptron whose weight matrices live inside a quantum circuit, and it composes naturally with variational layers. On three image-classification benchmarks, SBQE matches or outperforms both amplitude encoding and a width-matched classical network while requiring zero encoding gates and staying within NISQ depth budgets. These results suggest that the shot degree of freedom, already present in every quantum experiment, is a practical channel for data encoding on current hardware.

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The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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